

A 28-nm 9-kb SRAM Computing-in-Memory Macro With Segmented Charge Sharing for Multimode MAC Operations

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Abstract—Computing-in-memory (CIM) is a novel approach to solve the von Neumann bottleneck and improve energy efficiency and throughput. This article presents an SRAM-CIM macro based on segmented charge sharing to support multimode multiply-and-accumulate (MAC) operations, including binary weight network (BWN) MAC, ternary weight network (TWN) MAC, and multibit MAC operations. Signed 9-bit input MAC operations can be supported in BWN and TWN networks through multiple cycles. The multibit MAC operations are realized by the cooperation of two cell structures, which eliminates the need to convert negative numbers into complements and reduces the area and power consumption of the chip. Additionally, the proposed segmented charge-sharing scheme increases the speed of accumulation and improves the overall energy efficiency of the chip. The proposed 9-kb macro is implemented in 28-nm CMOS technology with an energy efficiency of 30.74 TOPS/W and an area efficiency of 1.15 TOPS/mm² in multibit MAC mode. At the system level, a ResNet-based model achieves an accuracy of 94.59% on the CIFAR-10 dataset and 74.64% on the CIFAR-100 dataset, demonstrating the effectiveness of the proposed CIM architecture for practical neural network inference. Compared to prior designs, it not only supports additional computing modes but also demonstrates notable improvements in both energy and area efficiency.

Index Terms—Computing-in-memory (CIM), multimode multiply-and-accumulate (MAC) operations, segmented charge sharing, SRAM.

I. INTRODUCTION

BREAKTHROUGHS in key technologies such as big data and artificial intelligence (AI) have fueled the rapid development of intelligent applications, exemplified by edge computing and smart living, that now characterize the modern digital era. Such applications often impose stringent demands on frequent memory access when executing tasks under the conventional von Neumann architecture [1].

However, the continuous shuttling of large volumes of data between memory and processing units incurs significant energy overhead and latency, which is widely recognized as the

von Neumann bottleneck. To address this fundamental limitation, computing-in-memory (CIM) has emerged as a promising paradigm by enabling parallel data processing within memory modules, thereby mitigating the von Neumann bottleneck and improving overall system efficiency [2], [3], [4], [5].

Convolutional neural networks (CNNs) form the backbone of modern AI applications, with convolution operations heavily relying on multiply-and-accumulate (MAC) computations. In recent years, silicon-based CIM research has primarily focused on SRAM-CIM architectures [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], as well as nonvolatile memory-based CIM (nvCIM) approaches [25], [26], [27] and eDRAM-based CIM approaches [28]. Although nvCIM schemes offer advantages in storage density, they still lag behind state-of-the-art SRAM-CIM solutions in terms of energy efficiency and computational accuracy for MAC operations.

In existing studies of SRAM-CIM, Si et al. [6] utilized local computing units and bit-detection circuits to implement unsigned input and weight MAC operations tailored for edge AI scenarios. Yu et al. [7] proposed an innovative SRAM-based 8T SRAM-CIM macro to achieve high energy efficiency under binary weight networks (BWNs). Ternary weight networks (TWNs), introduced in [8], quantize weights to “+1,” “0,” and “−1,” significantly improving computational accuracy compared to BWN. Building on this, Jain et al. [9] employed two 6T-SRAM cells to execute MAC operations for TWN, achieving even higher accuracy. With the continuous advancement of neural networks, there is a growing demand for better computational accuracy, higher energy efficiency, and greater processing flexibility.

In this work, we propose a novel SRAM-CIM macro with the following key contributions: the integration of computing units with array selection enables multimode operations, enhancing circuit flexibility; the inclusion of sign units and positive-negative charge computing cell (PNCC) enables MAC operations to handle both signed inputs and signed weights, thereby meeting the higher accuracy requirements for complex computational tasks; and the proposed segmented charge sharing accelerates charge sharing during accumulation without introducing additional MOM capacitors.

The remainder of this article is organized as follows. Section II discusses the challenges encountered in the design of high-precision SRAM-CIM macros. Section III details the proposed SRAM-CIM macro and its working principles. Sec-

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tion IV presents simulation results and performance analysis across different operation modes. Section V concludes this article with a summary of key findings.

II. CHALLENGES IN HIGH-PRECISION CIM DESIGN

Designing a unified SRAM-CIM architecture that simultaneously optimizes hardware utilization, energy efficiency, and computational precision remains challenging. In particular, supporting high-accuracy multimode MAC operations introduces several key issues. These issues include the limited flexibility of existing architectures in enabling multimode operations, inefficiencies in supporting signed inputs and signed weights for high-accuracy MAC computation, and performance bottlenecks in charge-domain accumulation. These challenges motivate the proposed multimode SRAM-CIM architecture described in Section III.

A. Limited Flexibility in Supporting Multimode MAC Operations

Most existing SRAM-CIM designs are well optimized for their targeted application scenarios; however, their circuit structures typically couple computing functionality with fixed array organizations, which limits the flexibility required for multimode MAC operations. For instance, the cascode current mirror-based inconsistency-free computing circuit in [10] improves the consistency and linearity of bitline charging and discharging rates, but its computation capability is limited to single-sign operations and relies on current-domain analog processing, which constrains energy efficiency. The work in [11] supports two operation modes, namely, single-sign MAC computation and grayscale image edge detection; however, its MAC operation depends on externally provided weight signs, limiting its applicability to more flexible computation modes.

Other designs, such as the two-way transpose SRAM-CIM macro in [12], offer high robustness and multibit MAC support for forward and backward propagation, yet are restricted to unsigned arithmetic. Similarly, the 6T SRAM-based CIM macro in [6] achieves low-latency 8-bit MAC operations but only supports unsigned MAC, with nMOS-connected local and global bitlines (GBLs) further increasing read/write latency.

These examples indicate that the lack of tight integration between computing units and array selection mechanisms fundamentally constrains circuit flexibility and the ability to support diverse MAC operations.

To address these limitations, this work tightly integrates the computing units with the array selection mechanisms, enabling a unified SRAM-CIM macro to flexibly support multiple MAC operation modes and thereby significantly enhancing the architectural reconfigurability and functional extensibility of the circuit.

B. Limited Support for Signed Inputs and Signed Weights in High-Accuracy MAC Operations

To enable signed MAC operations, many SRAM-CIM architectures employ dual-SRAM cell structures, which trade increased cell area and reduced array density for improved

signed computation capability and may also impact cell stability. For example, the dual-SRAM cell in [13] uses coupled capacitors to support variable-precision signed weights in the charge domain, but incurs substantial area overhead and low cell utilization.

An alternative approach combines local charge computing (LCC) with 6T SRAM cells to avoid duplicating storage; however, inherent limitations remain. Local bitlines are often connected to GBLs through nMOS devices, restricting pull-up capability and increasing read/write latency [14]. Moreover, signed computation based on two's-complement representation requires additional decoding logic, leading to further area overhead. Designs that avoid two's-complement arithmetic typically support either signed or unsigned operations exclusively.

As a result, existing architectures struggle to efficiently support MAC operations involving both signed inputs and signed weights, which are essential for achieving higher computational accuracy in complex tasks.

In this work, the inclusion of a sign cell (SC) and a PNCC enables MAC operations to directly handle both signed inputs and signed weights using sign-magnitude representation, without requiring extra two's-complement decoding. In addition, the SC improves the read/write latency of the LCC, enabling area-efficient and low-latency signed MAC operations while preserving array stability.

C. Efficiency Limitations in Charge-Domain Accumulation

Charge-sharing techniques have been extensively investigated to accelerate accumulation and improve energy efficiency in charge-domain SRAM-CIM designs. Some approaches introduce additional capacitors outside the memory array to perform weighting and accumulation of MAC results [15], [16], which leads to significant area overhead. Moreover, charge sharing involving large capacitances exacerbates accumulation and weighting latency, particularly for higher bit MAC operations. Although the method in [17] exploits bitline parasitic capacitance to enable charge sharing without introducing extra MOM capacitors, it partitions the GBL using nMOS switches, which degrades read/write speed, while the relatively small parasitic capacitance also increases sensitivity to noise. Furthermore, the charge-sharing mechanisms in [18] and [19] support accumulation only, and multibit weight processing still relies on subsequent shift-and-add operations.

These limitations reveal that conventional charge-sharing schemes are inefficient in accelerating charge sharing during accumulation, especially for multibit and high-precision MAC operations.

To address these issues, the proposed segmented charge-sharing scheme accelerates charge sharing during accumulation while utilizing only MOM capacitors overlaid above the memory cell layout, thereby improving overall efficiency without sacrificing computational accuracy.

III. PROPOSED SRAM CIM MACRO

A. Macro Architecture

Fig. 1(a) illustrates the overall architecture of the proposed SRAM-CIM macro, which supports multimode MAC

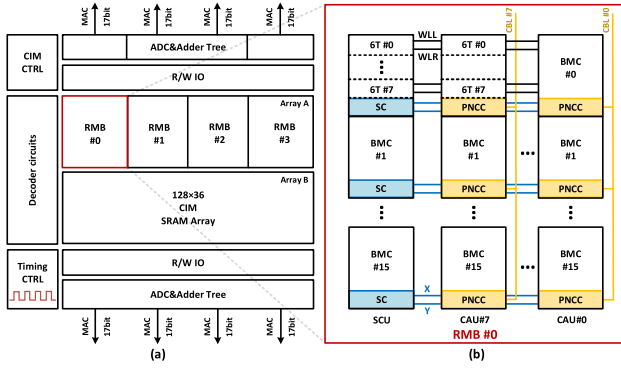


Fig. 1. (a) Architecture of the proposed CIM macro. (b) Structure of RMB.

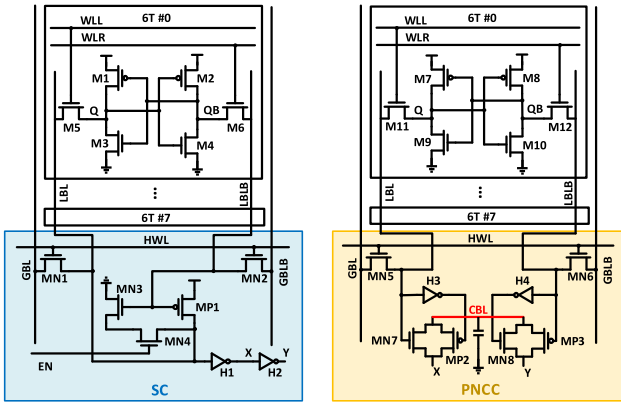


Fig. 2. Proposed SC and PNCC structures and the corresponding BMC integration with 6T SRAM.

operations using a segmented charge-sharing scheme. The key components include the memory array, precharge circuit, row and column decoder circuits, input driver, mode-switching logic, timing control unit, and output circuitry.

The memory array has a total capacity of 9 kb. Its basic storage element, bitwise multiplication cell (BMC), consists of eight independent-wordline 6T-SRAM cells along with either a PNCC or an SC. In Fig. 1(b), 16 SC-based BMCs stacked vertically constitute a signed computing unit (SCU), while 16 PNCC-based BMCs stacked vertically form a column accumulation unit (CAU). One SCU and eight CAUs are horizontally connected to construct a complete row MAC block (RMB). Four RMBs are arranged in rows to form an array, and two arrays (Arrays A and B) together constitute the complete 9-kb memory array. The macro supports two operation modes: storage mode and computing mode.

B. Proposed SC and PNCC Cell Structures and Read/Write Operations

Fig. 2 illustrates the proposed BMC structure, which consists of eight dual-wordline 6T-SRAM cells integrated column-wise together with either an SC or a PNCC. During the SC-based BMC write operation, the enable signal (EN) is set to a low level, enabling the SC write-assist path. To write a logical “1,” the write driver raises GBL to VDD while pulling GBLB down to ground. As depicted in Fig. 3(a), HWL is first

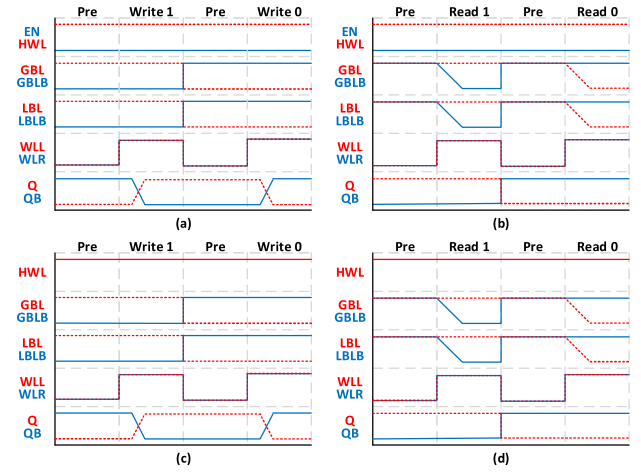


Fig. 3. Read and write timing diagrams of the proposed BMC. (a) Write operation of the SC-based BMC. (b) Read operation of the SC-based BMC. (c) Write operation of the PNCC-based BMC. (d) Read operation of the PNCC-based BMC.

asserted high to connect the GBLs to the local bitlines, and the local wordlines WLL and WLR of the target cell are then driven high to complete the write operation. In this case, the SC provides an effective write-assist mechanism to alleviate the threshold-voltage loss and improve the write speed.

Fig. 3(b) illustrates the read operation of the SC-based BMC when the selected cell stores a logical “0.” Before the read operation, the GBLs are precharged to VDD. Then, HWL is asserted high to connect the GBLs to the local bitlines, followed by asserting WLL and WLR of the selected row. As a result, GBL discharges, while GBLB remains at its precharged voltage. Finally, the sense amplifier (SA) detects the voltage difference between GBL and GBLB to read out the stored weight data from the selected 6T-SRAM cell.

Fig. 3(c) illustrates the write operation of the PNCC-based BMC. To write a weight of “0” to the selected 6T-SRAM cell, the write driver pulls GBL down to ground while driving GBLB up to VDD. As shown in Fig. 3(d), HWL is first asserted high to connect the GBLs to the local bitlines, and the local wordlines WLL and WLR are then driven high to complete the standard 6T-SRAM write process. Fig. 3(d) shows the read operation of the PNCC-based BMC, which follows a similar procedure to that of the SC-based BMC.

C. Circuit Implementation of Signed Multibit MAC Operations

Fig. 4 illustrates the timing diagrams of the proposed BMC during signed MAC operations, using a weight of +1 and input values of +1 and −3 as representative examples: 1) SC-based BMC and 2) PNCC-based BMC.

The signed inputs are fed into the computing unit through the local wordlines WLL and WLR. When the input sign is positive, WLL receives a positive pulse and WLR stays low; when the input sign is negative, WLR receives a positive pulse and WLL stays low. In this manner, the sign of the input is encoded by selecting one of the two input paths. These 2-bit inputs are encoded as pulsewidths, with “00,” “01,”

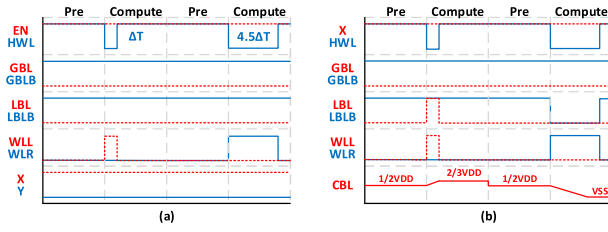


Fig. 4. Timing diagrams of the proposed BMC during signed MAC operations, illustrated with a weight of +1 and input values of +1 and -3. (a) SC-based BMC. (b) PNCC-based BMC.

TABLE I
TRUTH TABLE FOR SIGNED OPERATIONS

Input	WLL	WLR	Weight	Q	X	Y	Result	CBL
+	VDD	VSS	+1	1	VDD	VSS	+	Charge
			-1	1	VSS	VDD	-	Discharge
			0	0	\	\	0	Hold
-	VSS	VDD	+1	1	VDD	VSS	+	Charge
			-1	1	VSS	VDD	-	Discharge
			0	0	\	\	0	Hold
0	VSS	VSS	+1	1	VDD	VSS	0	Hold
			-1	1	VSS	VDD	0	Hold
			0	0	\	\	0	Hold

“10,” and “11” corresponding to $0\Delta T$, $1\Delta T$, $2\Delta T$, and $4.5\Delta T$, respectively. This pulsewidth modulation accounts for the nonlinear characteristics of capacitor charging and discharging, ensuring computational accuracy. As summarized in Table I, the sign bit of the weight is stored in the SC-based BMC, while the 1-bit weight magnitude is stored in the PNCC-based BMC.

1) *Precharge Phase*: Before computation starts, HWL is charged to connect local bitlines to GBLs. During this phase, HWL is asserted high to turn on the access transistors, through which LBL and LBLB are initialized to predefined voltage levels via the GBLs, thereby disabling the lower transmission gates in the PNCC. In the SC column, GBL, GBLB, LBL, and LBLB are precharged to a high level with the EN signal held high. In the PNCC column, CBL is precharged to $1/2VDD$, while LBL and GBL are discharged to ground to disable the left transmission gate, and LBLB and GBLB are charged to high to disable the right transmission gate. After precharging, HWL is discharged to disconnect the local and GBLs.

2) *Computation Phase*: WLL and WLR then transmit the input pulses, while the EN signal is set high. As shown in Fig. 2, in the SC unit, transistors MN3, MN4, and MP1 invert the signal QB transferred to LBLB and forward it to LBL. Consequently, regardless of whether the input pulse is applied through WLL or WLR, the signal appearing on LBL always represents the correct sign bit, which is subsequently delivered to the PNCC through H1 and H2 for further computation.

When WLL transmits a pulse, the left transmission gate in the PNCC is conditionally activated according to the stored weight value at node X. If the stored weight is zero, the transmission gate remains off and CBL retains its precharged state; if the stored weight is one, the transmission gate turns on, allowing node X to charge or discharge CBL. Similarly, when WLR transmits a pulse, the right transmission gate is activated according to the stored value at node Y. In this manner, the XNOR operation between the input sign and the weight sign is naturally completed, and the multiplication result is mapped

TABLE II
CORRESPONDENCE BETWEEN INPUT/WEIGHT VALUES
AND THE MAC RESULTS

Weight	Input		IN×W
BL/BLB	IN	WLL/WLR	CBL
+1/-1	+11	$4.5\Delta T$	VDD or 0
	+10	$2\Delta T$	$5/6VDD$ or $1/6VDD$
	+01	$1\Delta T$	$2/3VDD$ or $1/3VDD$
	00	0	$1/2VDD$
	-01	$1\Delta T$	$1/3VDD$ or $2/3VDD$
	-10	$2\Delta T$	$1/6VDD$ or $5/6VDD$
	-11	$4.5\Delta T$	0 or VDD
0	X	X	$1/2VDD$

to the voltage variation on CBL. After the pulse ends, HWL reconnects the local bitlines to the GBLs and disables the transmission gates.

After the multiplication computing, the segmented charge-sharing scheme (as described in Section III-D) is used to redistribute the charge on CBL in two steps, completing the MAC operation for 16 2-bit inputs and 1-bit weights. This configuration enables the weighted accumulation of single-bit weighted MAC results from each column according to their respective weights, yielding a three-bit weighted MAC result that is subsequently read out by the ADC. The same principle applies to the mid-3 bits and the high-2 bits. The outputs from the ADCs are then shifted and accumulated through an adder tree to generate the MAC result for 9-bit weights (including one sign bit) and 3-bit inputs (including one sign bit). This weighting approach not only reduces the number of required ADCs, thereby saving area, but also decreases the number of shift-and-accumulate operations, enhancing computational speed. Finally, the 9-bit signed input is partitioned into four segments of 3-bit signed inputs, and the aforementioned process is repeated four times to obtain four MAC results. These results are then shifted and accumulated according to their weights to produce the final MAC result for 9-bit signed inputs and 9-bit signed weights.

It is worth noting that the voltage of the capacitor does not vary linearly with time during charging or discharging through a transmission gate. Therefore, the 2-bit inputs are encoded using pulsewidths of 0, $1\Delta T$, $2\Delta T$, and $4.5\Delta T$, as listed in Table II. The selection of ΔT is inherently coupled with the intended array dimensions and the characteristics of the pulse delivery network and should be tailored accordingly in practical implementations.

At the beginning of the discharge, both MOS transistors in the transmission gate operate in the saturation region, resulting in an approximately constant current and a near-linear voltage decrease on the capacitor. As the capacitor voltage decreases, the pMOS device gradually turns off and the nMOS device enters the linear region, leading to a reduced current and nonlinear voltage behavior. To compensate for this effect, the maximum pulsewidth corresponding to the largest input magnitude is intentionally extended, ensuring that the four input levels map to approximately linear voltage levels and thereby guaranteeing the correctness of subsequent accumulation and weighting operations.

Fig. 5 illustrates the mapping of the proposed CIM module for supporting 9-bit signed multiplication in neural networks. In this configuration, both the input activations and weights

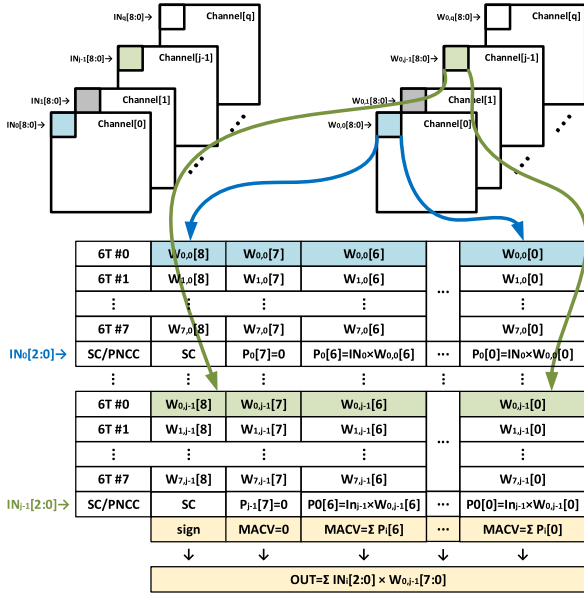


Fig. 5. Mapping of the proposed CIM module for supporting 9-bit signed MAC in neural networks.

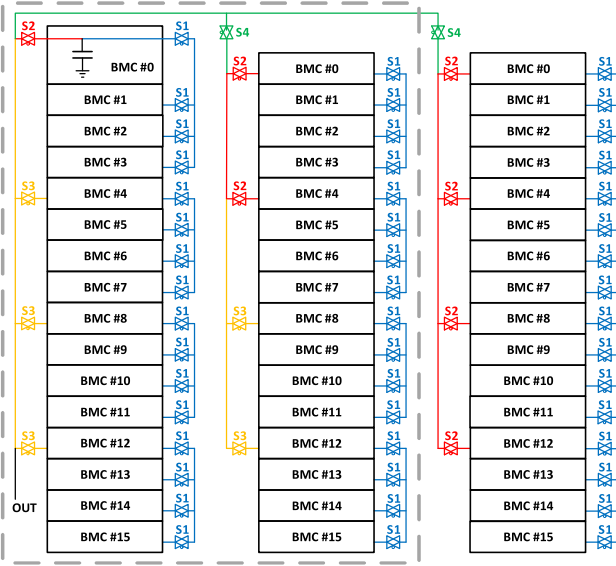


Fig. 6. Transistor-level implementation of the proposed multimode reconfigurable segmented charge-sharing architecture.

are quantized to 9-bit signed values. The proposed mapping strategy aligns with the CIM operation flow, enabling efficient accumulation of partial products within the array and subsequent digital processing through the ADC and adder tree.

D. Multimode Reconfigurable Segmented Charge Accumulation Circuit

Fig. 6 presents the transistor-level circuit implementation of the proposed multimode reconfigurable segmented charge-sharing architecture. The proposed circuit is capable of not only accumulating the computation results generated by BMC composed of PNCC, but also performing intercolumn weighted accumulation of stored weights. By appropriately

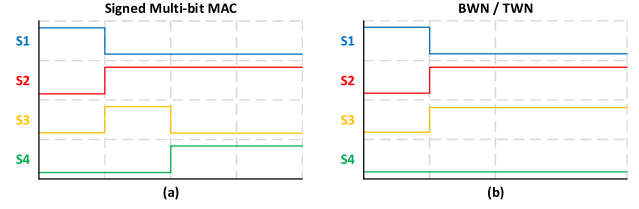


Fig. 7. Timing diagrams of the proposed circuit for different computing modes. (a) Signed multi-bit MAC operations. (b) MAC operations of BWN and TWN.

configuring four control signals, the same circuit can support multiple computing modes, including multi-bit MAC operations as well as the MAC operations of BWN and TWN, thereby providing high computational flexibility with minimal hardware overhead.

As illustrated in Fig. 7(a), during signed multi-bit MAC operation, switch S1 is first turned on to group every four BMCs as a basic local accumulation unit. In this phase, the computation results of four adjacent BMCs are locally accumulated through charge sharing, forming the first-stage partial sum. Subsequently, S1 is turned off and switches S2 and S3 are simultaneously enabled. In this configuration, the first partial sum from each local accumulation unit is selected and extracted to participate in the global accumulation process, as mathematically described in the following equation:

$$V_{ac} = \frac{1}{4} \sum_{i=0}^{15} CV_i = \frac{\sum_{i=0}^{15} V_i}{16}. \quad (1)$$

The key advantage of this segmented charge-sharing strategy lies in the fact that only four capacitors are involved in charge sharing at each accumulation step. Compared with conventional schemes in which 16 capacitors simultaneously participate in charge sharing, the proposed approach significantly reduces the effective capacitive load. This reduction leads to a substantial improvement in charge-sharing speed, especially when relatively large accumulation capacitors are employed to enhance signal resolution and noise margin.

After the global accumulation of partial sums, switch S3 is turned off and switch S4 is enabled. At this stage, intercolumn 3-bit weight scaling is realized by exploiting the capacitor participation ratios of 1, 2, and 4 during charge sharing, as formulated in (2). Through this capacitor ratio-based charge redistribution, weighted accumulation across columns is efficiently implemented without requiring additional digital multipliers or complex analog circuits

$$V_{OUT} = \frac{1}{7C} (CV_{ac1} + 2CV_{ac2} + 4CV_{ac3}). \quad (2)$$

As shown in Fig. 7(b), during the MAC operations of BWN and TWN, the accumulation process follows the same procedure as that of MAC operation. However, since the weight is limited to 1 bit in these modes, intercolumn weighting is no longer required. Once accumulation is completed, the final result can be directly read out through the ADC, resulting in a simplified operation flow and reduced latency for low-precision neural network inference.

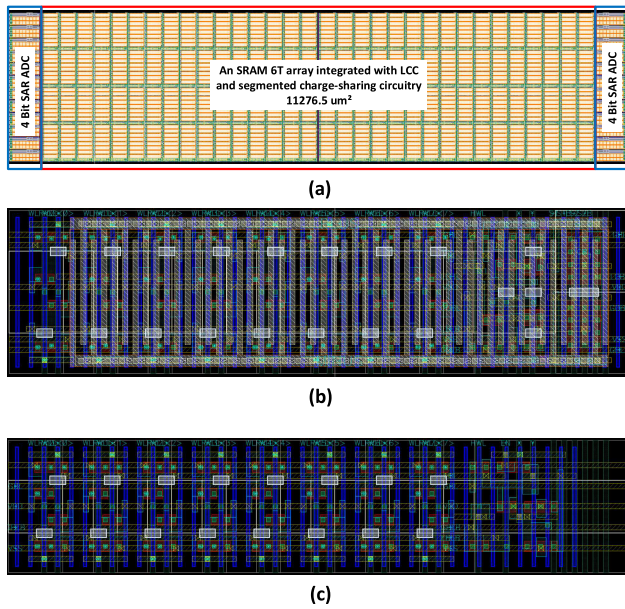


Fig. 8. (a) Layout of the proposed 9-kb SRAM CIM macro. (b) Layout of BMC with PNCC. (c) Layout of BMC with SC.

IV. SIMULATION RESULTS AND ANALYSIS

The proposed SRAM macro is implemented using a 28-nm CMOS process with a standard supply voltage (VDD) of 0.9 V. In this design, the minimum transistor size is employed for the 6T-SRAM cells, and the overall array size is 9 kb. Fig. 8(a)–(c) presents the layout of the proposed 9-kb SRAM CIM macro, the layout of BMC with PNCC, and the layout of BMC with SC for this design, respectively. The total area of the 9-kb array integrating the 6T-SRAM array, the LCC circuitry, and the segmented charge-sharing accumulation circuit is 11276.5 μm^2 .

Noise tolerance is a critical performance metric for SRAM, as it indicates the stability of the memory under various conditions. Fig. 9(a) and (b) shows the hold and write noise margins of the proposed circuit across different temperatures and process corners, demonstrating that the circuit maintains good stability under these conditions.

Since the sign column employs a dedicated sign calculation unit, it enhances the write operation and improves the overall write speed of the circuit. As shown in Fig. 9(c), the read and write delays of the SC unit were evaluated under different temperatures and supply voltages. At 27 °C with a supply voltage of 0.9 V, the write delay is as low as 24.8 ps. In contrast, the write capability of the uncompensated computing column is illustrated in Fig. 9(d). At 27 °C and a supply voltage of 0.9 V, its write delay increases to 33.5 ps. A comprehensive comparison across different process corners and temperatures, as shown in Fig. 9(c) and (d), demonstrates that the proposed sign calculation unit achieves a significant improvement in write performance.

To verify the stability of the computation process, Fig. 9(e) and (f) presents the results of 2000 Monte Carlo simulations of the bitline CBL at 27 °C under the TT process corner, for calculations of +1 and −1. After 2000 simulations, the mean

value of $+\Delta V$ is 597.566 mV with a variance of 12.575 mV, while the mean value of $-\Delta V$ is 301.375 mV with a variance of 18.295 mV at a supply voltage of 0.9 V. These results indicate that the computation remains stable under typical operating conditions. To perform MAC operations, different output voltages correspond to different computational results. The linearity of the MAC operation was analyzed using the voltage difference method and the integral nonlinearity (INL). The expressions for ΔV and INL are defined as follows:

$$\Delta V = \frac{V_{max} - V_{min}}{n} \quad (3)$$

$$INL = \left| \frac{V_{cal} - V_{min}}{\Delta V} - m \right| \quad (4)$$

where n denotes the maximum MAC result, V_{max} represents the discharge voltage for the maximum result, V_{min} represents the discharge for the minimum result, m is the actual MAC result, and V_{cal} is the actual calculated discharge voltage. A smaller INL value indicates a more reliable and linear MAC operation.

As shown in Fig. 10(a), the blue curve represents the actual discharge voltage V_{cal} , while the red curve shows the ideal values derived from the INL calculation. With a single bitline discharge and a simulated maximum MAC result of 48, the voltage difference between two adjacent calculation levels is approximately 9.375 mV. Additionally, the INL value remains below 3 LSBs (approximately 0.9 LSB), demonstrating good linearity for the MAC computation.

During the accumulation process, the proposed design employs a segmented charge-sharing scheme, which significantly accelerates charge redistribution. As illustrated in Fig. 10(b), the figure compares the computation speeds of a single column of cells under identical total accumulation capacitance and identical number of drivers, for the cases with and without segmented charge sharing. When the segmented scheme is applied, the slowest sharing time (corresponding to a MAC result of “1”) is approximately 1.5 ns, whereas without segmentation, the sharing time extends to about 12 ns. This indicates that the segmented charge-sharing approach reduces the accumulation time by roughly a factor of eight, thereby substantially shortening the overall computation time in the computing mode.

The simulation of this circuit involves measuring the average current during the arithmetic operation, calculating the dynamic power by multiplying the average current by the supply voltage, and then multiplying by the duration of the MAC operation to obtain the total energy consumption. Fig. 10(c) and (d) presents the simulated power consumption and computational efficiency for different supply voltages and process corners at 27 °C in MAC mode with signed input weights. The results show that at VDD = 0.7 V, the lowest energy consumption is 9.52 fJ/bit with a corresponding computational efficiency of 105.04 TOPS/W. Under the slow–slow (SS) process corner at VDD = 0.9 V, the lowest energy consumption is 24.38 fJ/bit with a computational efficiency of 41.02 TOPS/W.

Fig. 10(e) presents a comparison of the simulated energy consumption between the proposed design and prior works

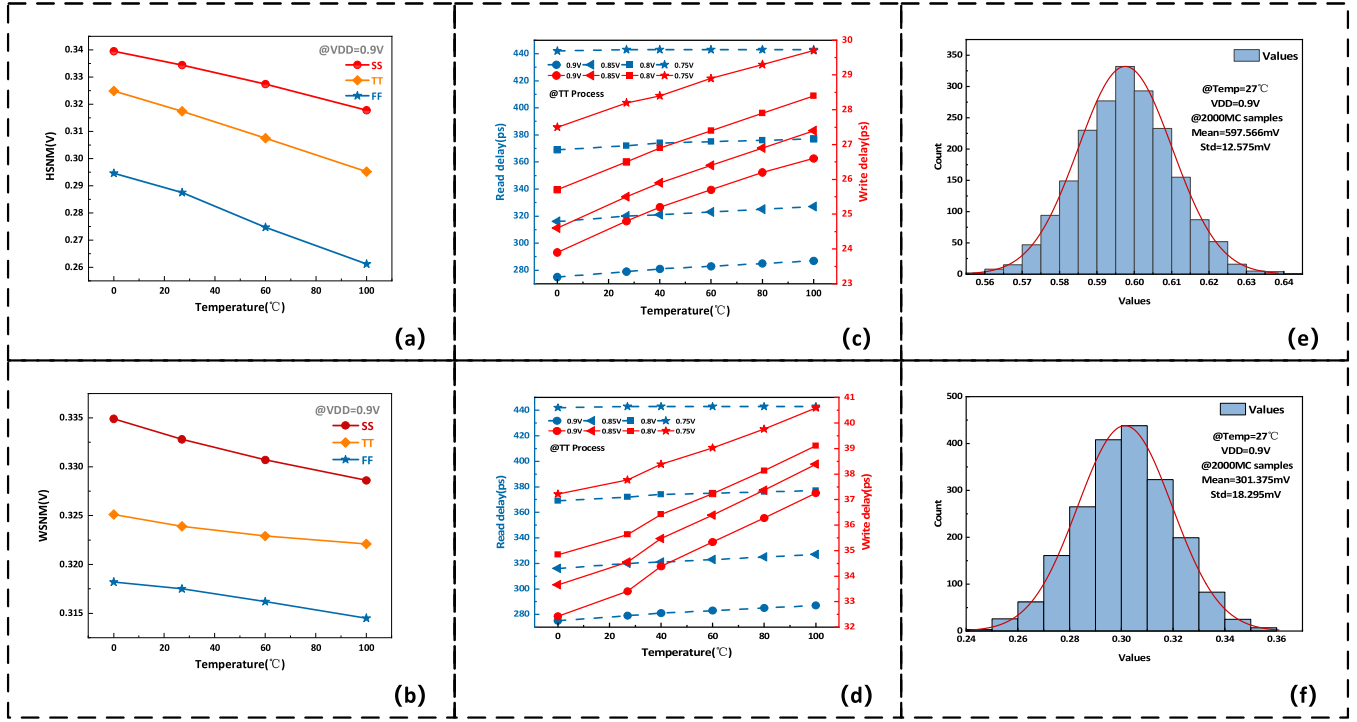


Fig. 9. (a) Hold static noise margin. (b) Write static noise margin static. (c) Read and write delays of SCs at different temperatures and different processes. (d) Read and write delays of PNCC at different temperatures and different processes. (e) Monte Carlo simulation with calculated $+\Delta V$. (f) Monte Carlo simulation with calculated $-\Delta V$.

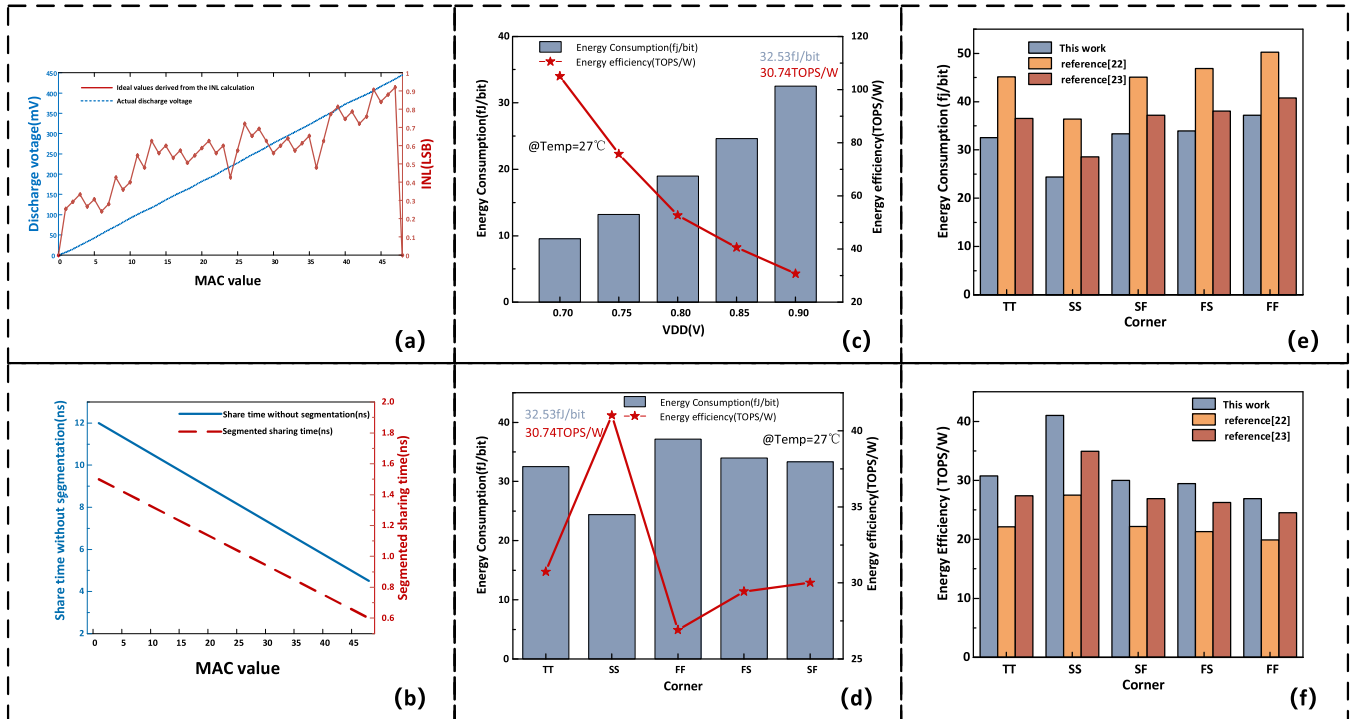


Fig. 10. (a) Differential voltage and linearity of bitline voltage in MAC operation. (b) Comparison of computing time using the segmented charge-sharing scheme and not using the segmented sharing scheme. (c) Energy consumption and energy efficiency for different VDDs. (d) Energy consumption and energy efficiency for different process corners. (e) Comparison of energy consumption across different process corners at 27 °C. (f) Comparison of energy efficiency across different process corners at 27 °C.

[22], [23] under typical process corners at room temperature (27 °C). Simulation results demonstrate that the proposed scheme achieves energy efficiencies of 32.53, 24.38, 37.16, 33.97, and 33.31 fJ/bit under process corners. At 27 °C under

TABLE III
COMPARISON WITH PRIOR SRAM-CIM WORKS

	<i>JSSC'21</i> ^[14]	<i>JSSC'23</i> ^[17]	<i>ISSCC'23</i> ^[20]	<i>VLSI'24</i> ^[21]	<i>ISSCC'23</i> ^[22]	<i>ISSCC'22</i> ^[23]	<i>This work</i>
Technology(nm)	28	28	28	28	22	28	28
Array size(Kb)	64	384	9	16	128	32	9
Supply voltage(V)	0.7-0.9	0.7-0.9	0.6-0.9	0.65-0.9	0.72-0.82	0.8	0.7-0.9
Input(bit)	8	8	8	8	8	8	9
Weight(bit)	8	8	8	8	8	8	9
Access time(ns)	8.4	7.2	6	5.6	22.16	24	5
Energy Efficiency (TOPS/W)	11.54-16.63	15.02-22.75	14.24-33.44	13.31-40.15	16.02-21.38	10.21-35.55	26.91-41.02*
Area Efficiency (TOPS/mm²)	-	0.456	1.125	1.111	1.44	-	1.15*
Hardware-Aware Accuracy (With CIM Non-Idealities)	92.02%** 67.89%***	-	-	-	-	-	94.59%** 74.64%***

*Under 9 bit input and 9 bit weight conditions **ResNet on CIFAR-10 with circuit non-idealities considered ***ResNet on CIFAR-100 with circuit non-idealities considered

the TT process corner, the proposed design reduces energy consumption by 28.4% compared to [22] (45.13 fJ/bit) and by 10.93% compared to [23] (36.52 fJ/bit).

Fig. 10(f) presents a comparison of energy efficiency under process corner simulations at room temperature (27 °C) between the proposed design and prior works [22], [23]. Simulation results demonstrate that the proposed scheme achieves energy efficiencies of 30.74, 41.02, 26.91, 29.44, and 30.02 TOPS/W under TT/SS/FF/FS/SF process corners, respectively. At 27 °C under the TT process corner, the proposed design exhibits a 43.78% improvement compared to [22] (21.38 TOPS/W) and a 12.27% enhancement over [23] (27.38 TOPS/W).

Table III presents a comparison between the proposed CIM macro and other existing techniques. In terms of access time, which is defined based on the intrinsic computing latency of the CIM macro, it is significantly lower than the designs reported in [22] and [23] and also shows a certain improvement compared to the high-performance designs in [20] and [21]. Furthermore, in terms of energy efficiency, both the maximum and minimum values of this design exhibit substantial improvements over other designs listed in the table.

Compared with previous works, the circuit structure proposed in this article employs dedicated sign and trailing columns, enabling MAC operations with signed inputs and weights. This design allows the computing macro to support a broader range of network types while maintaining the computational accuracy required for complex scenarios. Moreover, the use of segmented charge sharing during accumulation significantly accelerates charge redistribution and reduces the overall computation time.

Although the proposed design may face potential risks related to HWL sensitivity and segmented charge-sharing

operations, system-level evaluations that incorporate circuit nonidealities demonstrate strong robustness. Specifically, a ResNet-based model achieves an accuracy of 94.59% on the CIFAR-10 dataset and 74.64% on the CIFAR-100 dataset.

V. CONCLUSION

This work presents a 28-nm SRAM-CIM circuit designed to support multimode MAC operations. To eliminate the need for negative number complement conversion and effectively reduce area and power consumption, SC and PNCC are proposed to enable multibit signed MAC functionality. Furthermore, a segmented charge-sharing scheme is adopted to overcome the limitations of conventional charge-sharing schemes, significantly accelerating the accumulation process while enhancing overall computational speed and energy efficiency. At the system level, a ResNet-based model achieves an accuracy of 94.59% on the CIFAR-10 dataset and 74.64% on the CIFAR-100 dataset, demonstrating the effectiveness of the proposed architecture for practical neural network inference. Simulation results demonstrate that the 9-kb macro achieves an energy efficiency of 30.74 TOPS/W and an area efficiency of 1.15 TOPS/mm² in multibit signed MAC mode, providing an efficient CIM solution for multibit signed MAC operations.

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